

Monte Carlo Dropout for Object Detection on Point Clouds

Guided Research

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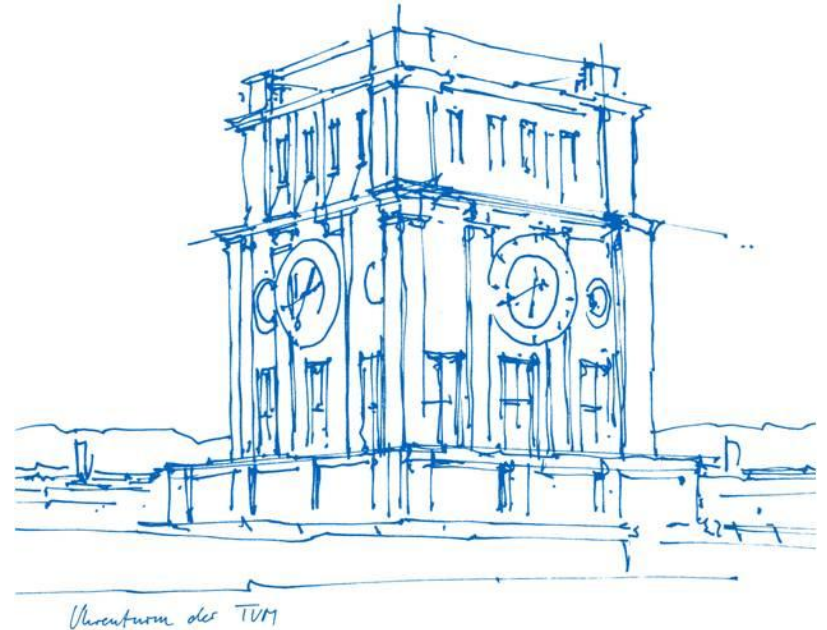
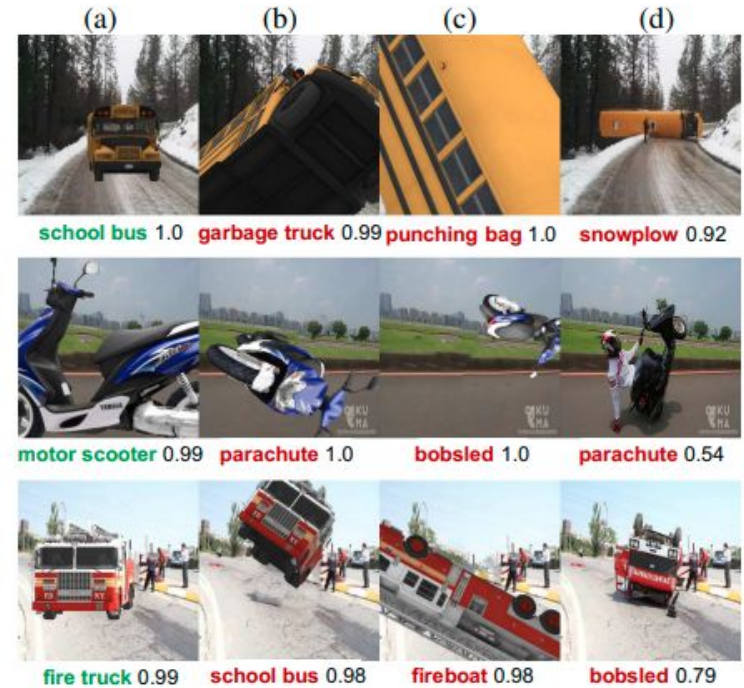


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Motivation

Deep learning methods can make wrong but very confident predictions



[Alcorn et al. 2018]

Motivation

Deep learning methods can make wrong but very confident predictions

Google apologizes for algorithm mistakenly calling black people 'gorillas'

Image source [CNet, 2015]

Tesla driver dies in first fatal crash while using autopilot mode

The autopilot sensors on the Model S failed to distinguish a white tractor-trailer crossing the highway against a bright sky

Image source [The Guardian, 2016]

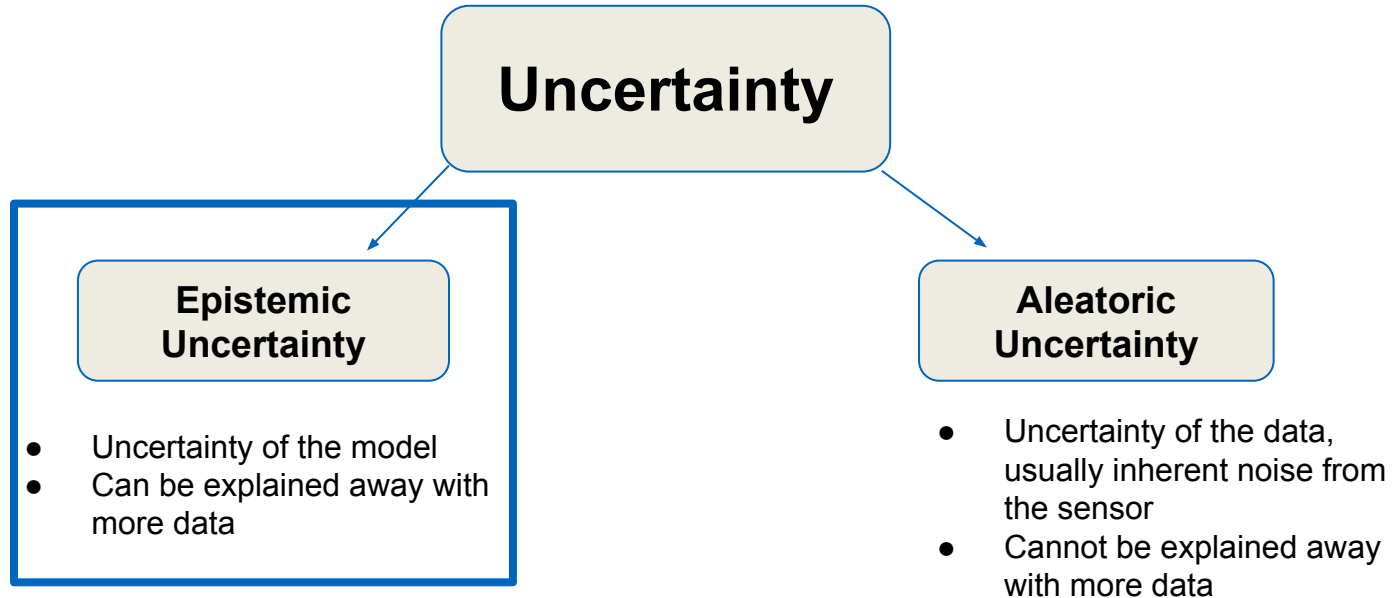
Motivation

- Leveraging uncertainty information for 3D scene understanding is an underexplored area
- Overconfident predictions can cause catastrophic consequences, it is critical in real life scenarios to be aware of the uncertain situations.
- Train models that are able recognize when they are likely to make mistakes.

Related Work: 3D Scene Understanding

- Deep Hough Voting for 3D Object Detection in Point Clouds [Qi et al, 2019]
- Mix3D: Out-of-Context Data Augmentation for 3D Scenes [Nekrasov et al, 2021]
- RfD-Net: Point Scene Understanding by Semantic Instance Reconstruction[Nie et al, 2021]
- TransformerFusion: Monocular RGB Scene Reconstruction using Transformers [Božič et al., 2021]

Related Work: Uncertainty



Related Work: Epistemic Uncertainty in Deep Learning

- In order to capture epistemic uncertainty in a neural network, we put a prior distribution (usually Gaussian) over its weights and instead of optimizing weights directly, we average over all possible weights (marginalization) during inference, these models are referred as Bayesian neural networks. [Kendall et al. 2017]

Related Work: Bayesian Neural Networks (BNN)



- Performing inference in BNN's is done by applying the Bayes' Rule for evaluating the posterior probability $p(\mathbf{W} \mid \mathbf{X}, \mathbf{Y}) = p(\mathbf{Y} \mid \mathbf{X}, \mathbf{W})p(\mathbf{W})/p(\mathbf{Y} \mid \mathbf{X})$
- This is nice to formulate, but often intractable to do in practice, because we need to evaluate marginal probability $p(Y|X)$ over all the parameters.
- Hence, in order to perform inference in BNN's we need to approximate true posterior.

Related Work: Monte Carlo (MC) Dropout [3]

- Existing approximation methods follow the strategy of defining a simple distribution that fits into the posterior distribution, and then optimizing the simple distribution instead of the true distribution [Kendall et al, 2017].
- [Gal et al. 2016] Has shown that performing dropout at test time is equivalent to approximating the posterior distribution with a Gaussian which minimized the distance between the distributions, and that we can easily compute the uncertainty of complex and deep models without having to tackle an intractable problem.

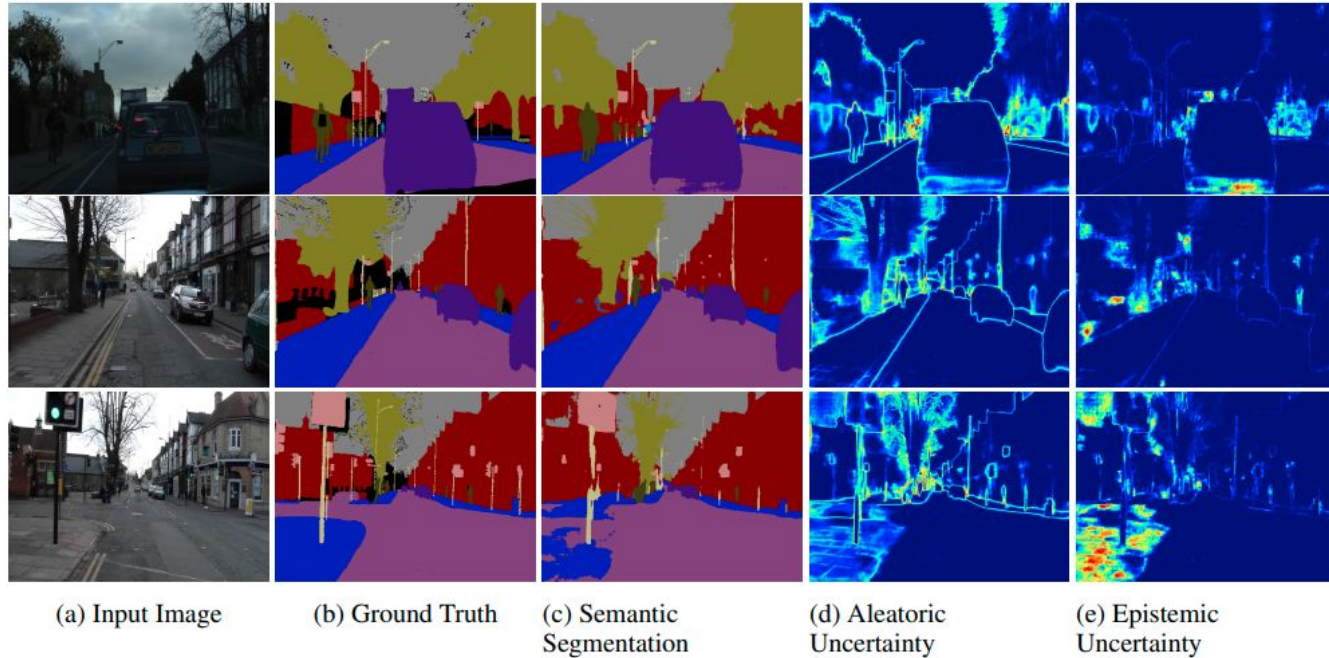
Related Work: Epistemic Uncertainty in Classification

- Expected Entropy
 - High when there's ambiguity in the decisions
 - MC approximation: Expected Entropy $\sim \frac{1}{T} \sum_{t=1}^t (H(f_t^W(x)))$

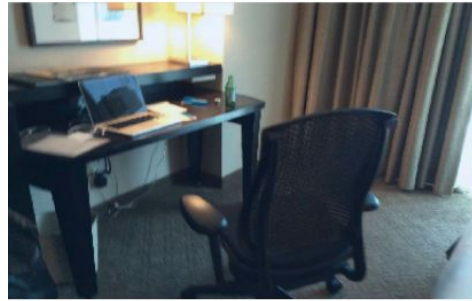
Related Work: Epistemic Uncertainty in Classification

- Predictive Entropy
 - Entropy of the expected prediction
 - High when there's ambiguity
 - High when the predictions far away from the data
- Mutual Information
 - $\text{Mutual Information} = \text{Predictive Entropy} - \text{Expected Entropy}$
 - Only high when far away from data

Related Work: What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?



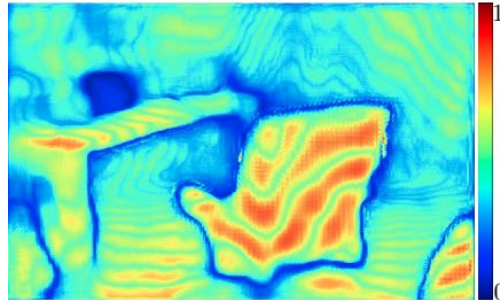
Related Work: Neural RGB→D Sensing: Depth and Uncertainty from a Video Camera



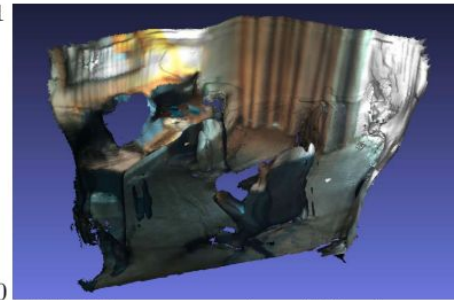
Input frame



Estimated depth



Confidence



3D Recon. using 30 views

Related Work: Neural RGB→D Sensing: Depth and Uncertainty from a Video Camera

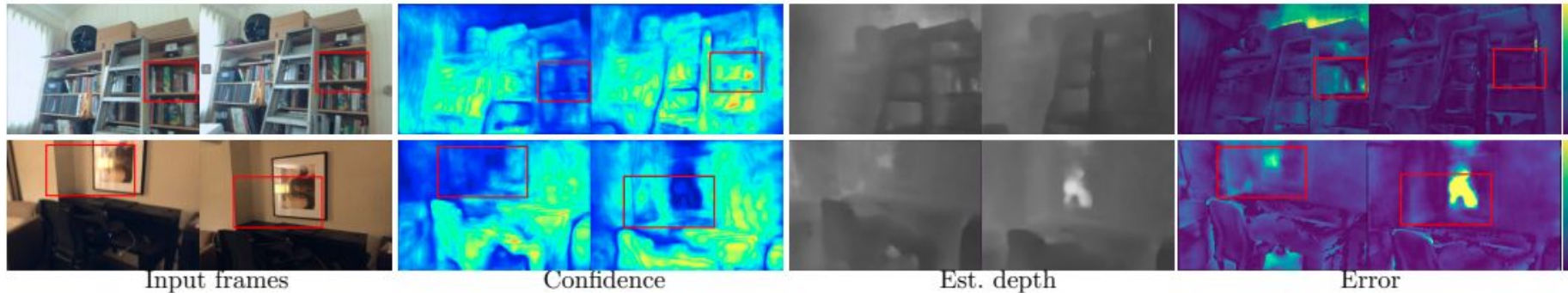


Image source [Liu et al, 2019]

Related Work: D3VO: Deep Depth, Deep Pose and Deep Uncertainty for Monocular Visual Odometry

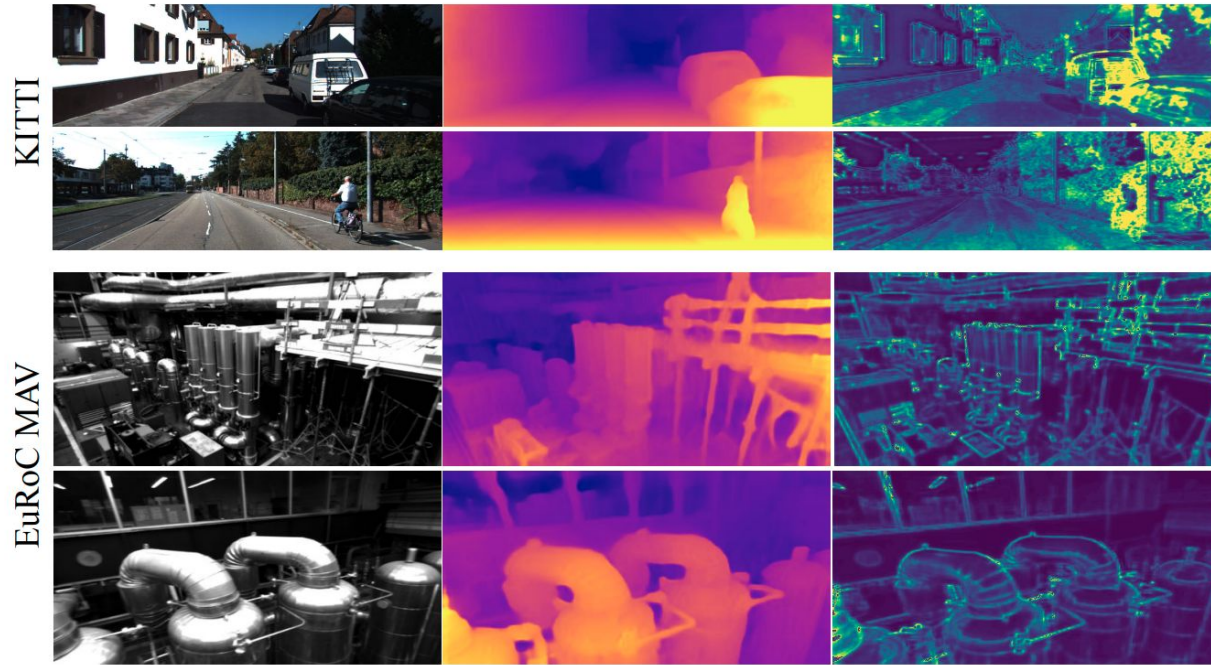
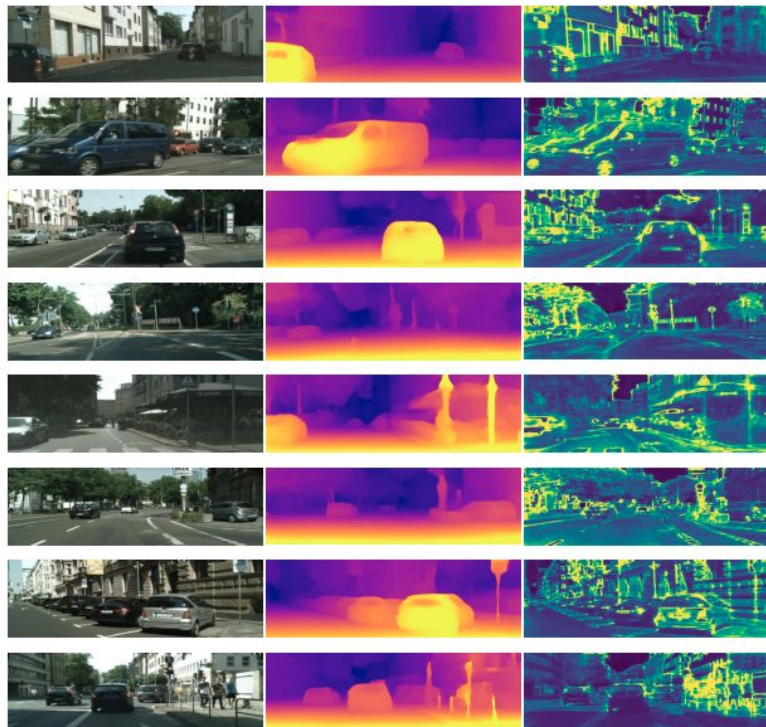
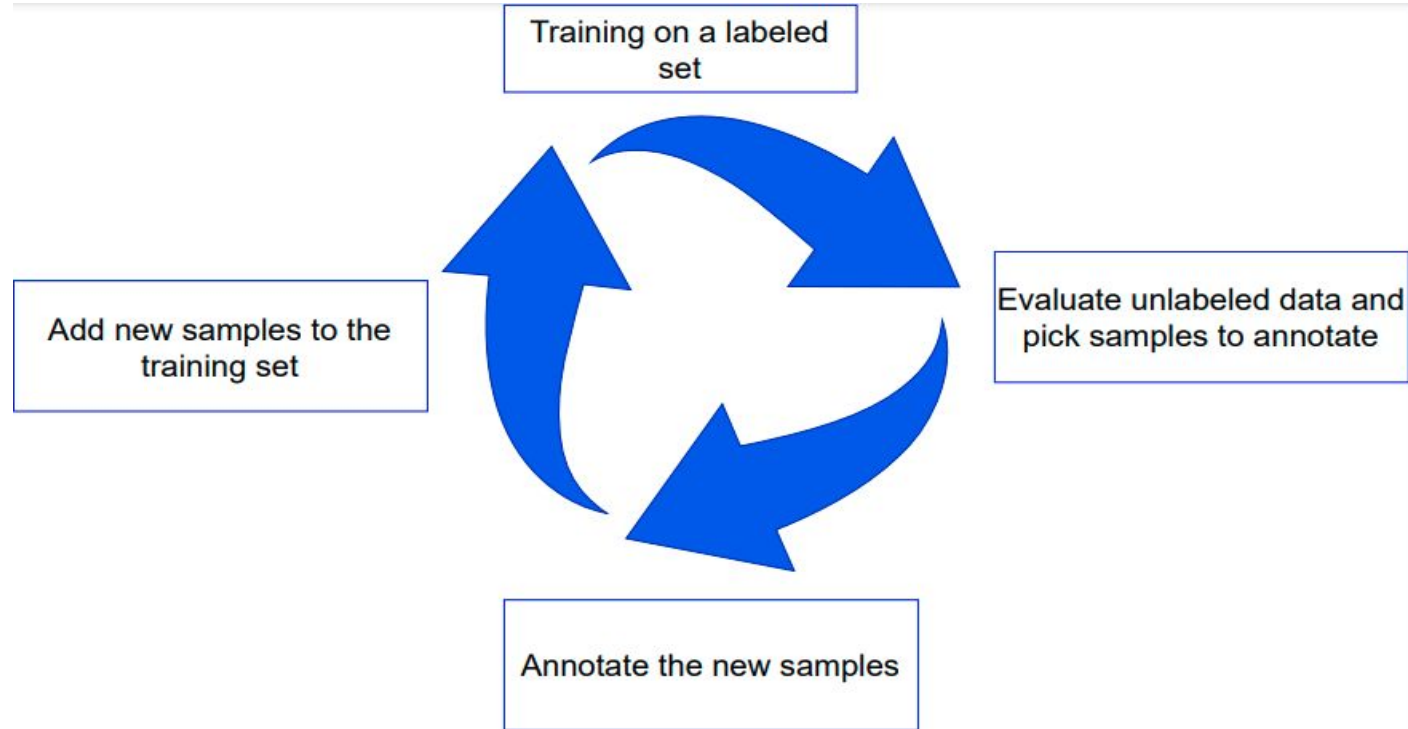


Image source [Yang et al, 2020]

Related Work: D3VO: Deep Depth, Deep Pose and Deep Uncertainty for Monocular Visual Odometry



Related Work: Active Learning



Related Work: Learning Loss for Active Learning

[Yoo et al, 2019]

- Learning to predict what would the loss value be for an unlabeled input
- Selecting data where this prediction suggests that model is mistaken

Related Work: Bayesian Active Learning by Disagreement (BALD) [Houlsby et al, 2011]

- Sample the data with a Bayesian model
- Pick the samples with largest mutual information
- Add it to the dataset

Related Work: BatchBALD: Efficient and Diverse

Batch Acquisition for Deep Bayesian Active Learning

[Kirsch et al, 2019]

- Similar to BALD, it is enforcing batch-aware selections

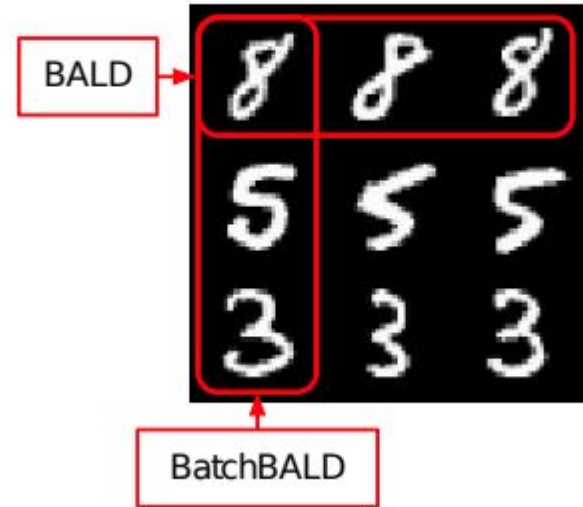


Image source [Kirsch et al, 2019]

Method: Object Detection with VoteNet [Qi et al, 2019]

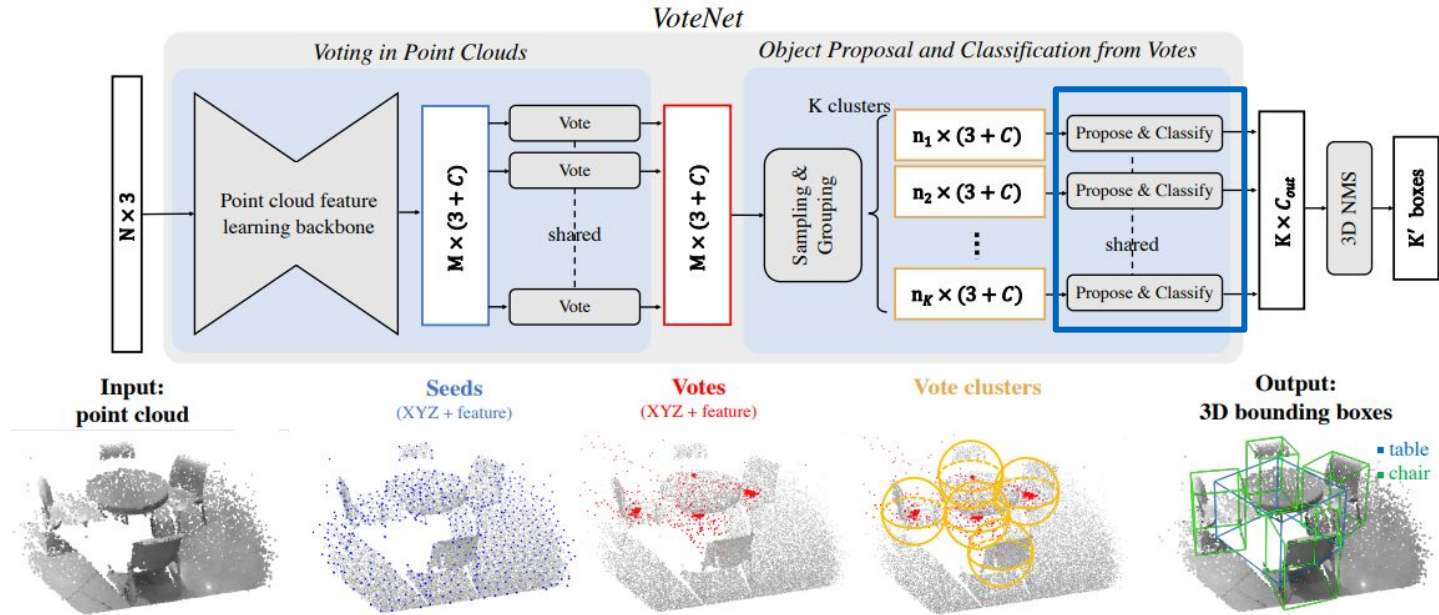
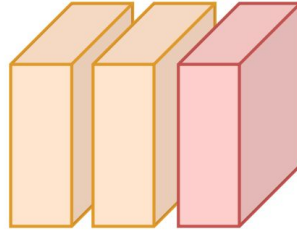


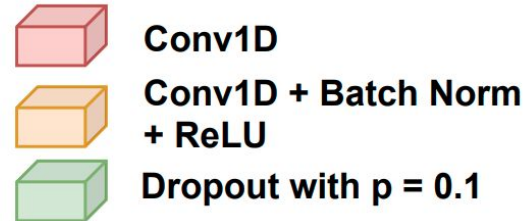
Image source [Qi et al, 2019]

Method: Extending VoteNet

Original Proposal Module

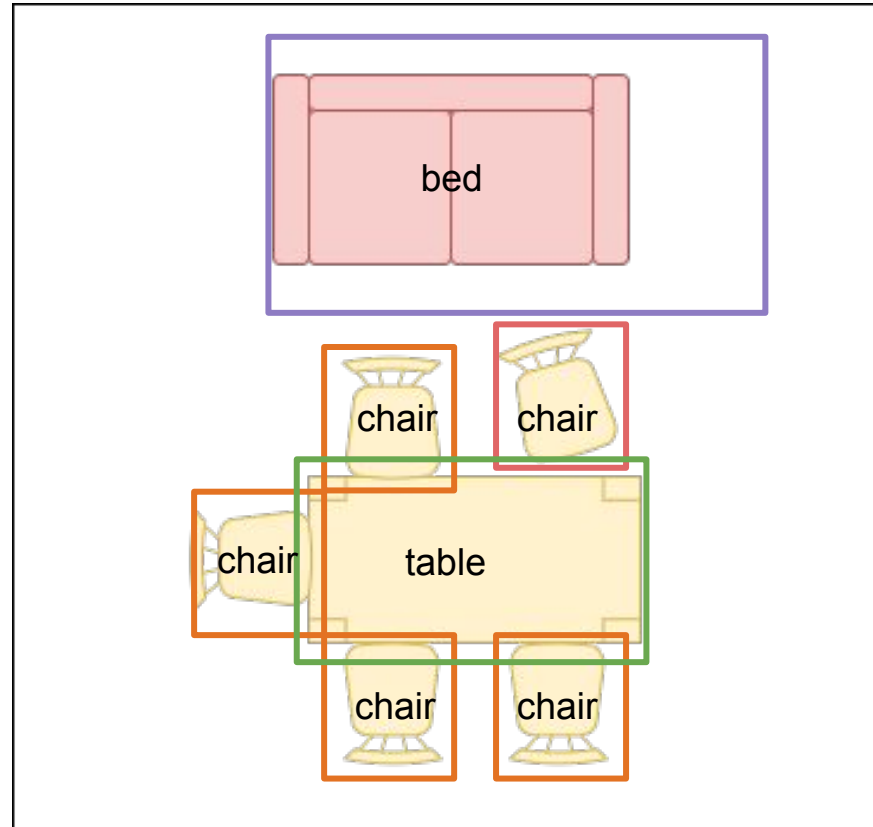


Extended Proposal Module



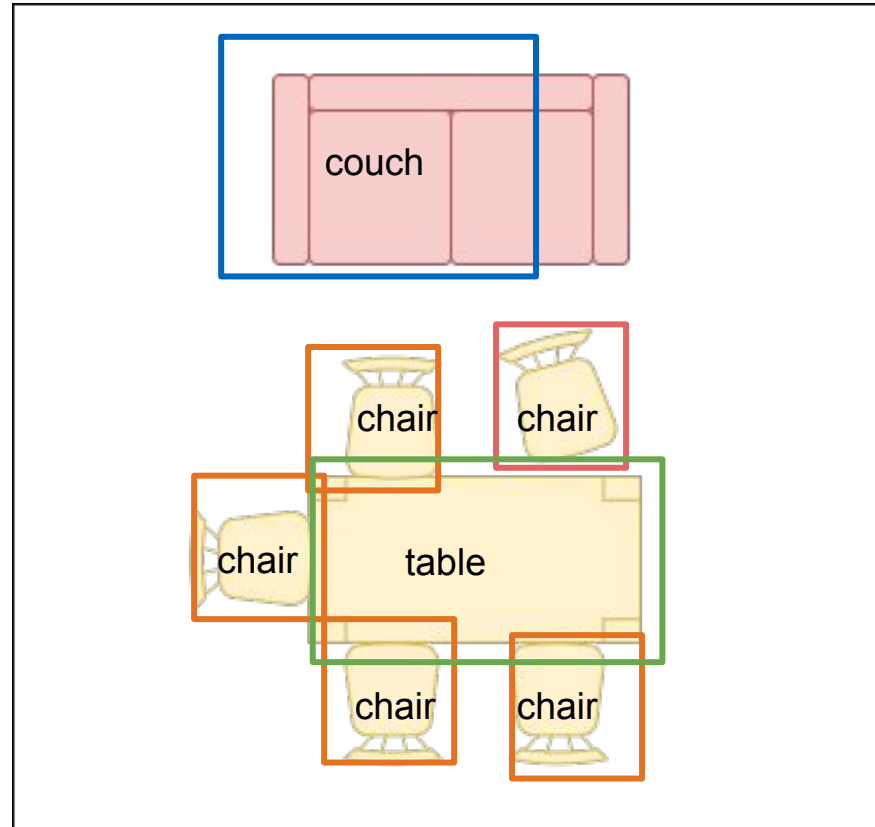
Method: Monte Carlo Dropout

Perform T forward passes with dropouts enabled



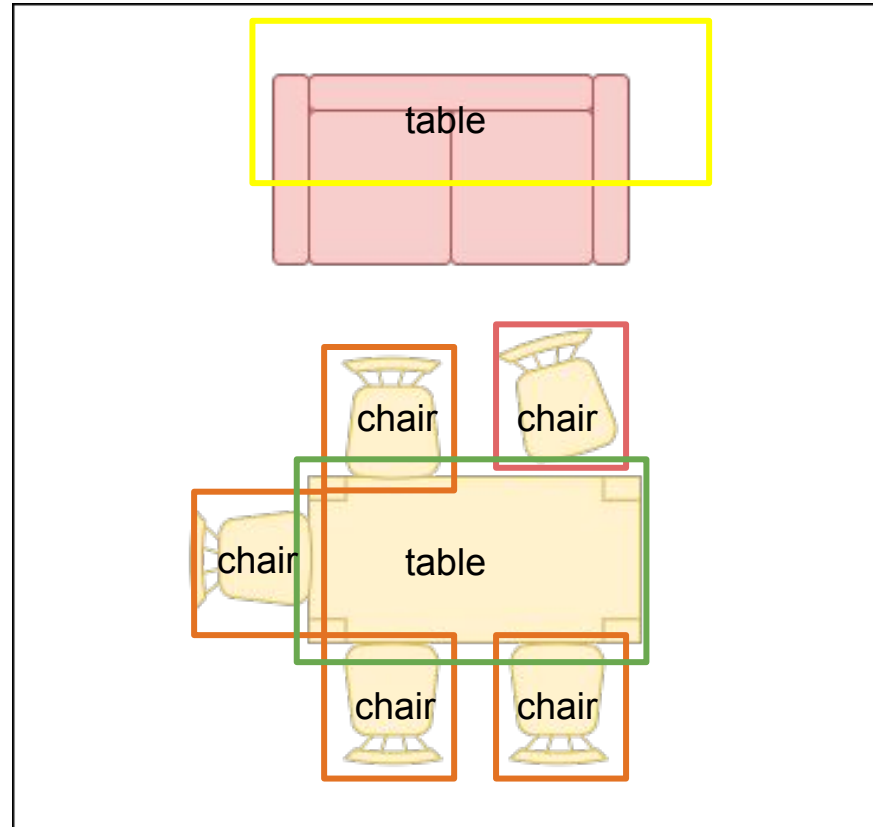
Method: Monte Carlo Dropout

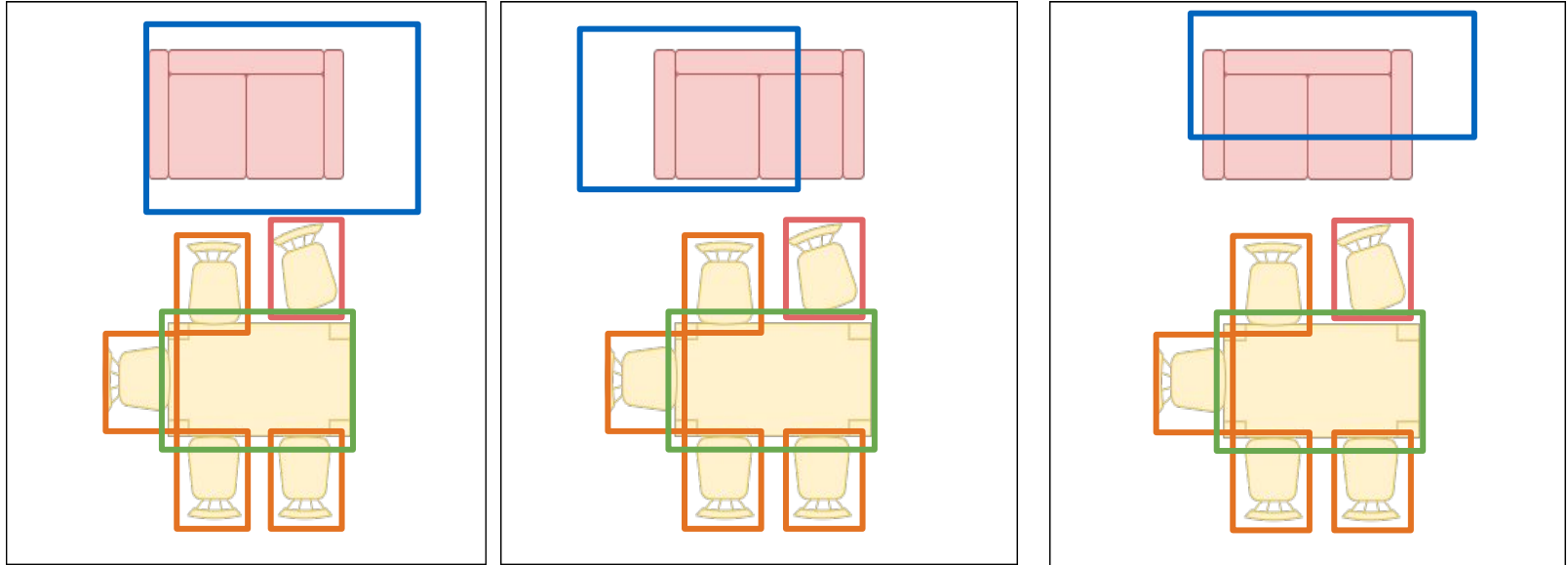
Perform T forward passes with dropouts enabled



Method: Monte Carlo Dropout

Perform T forward passes with dropouts enabled

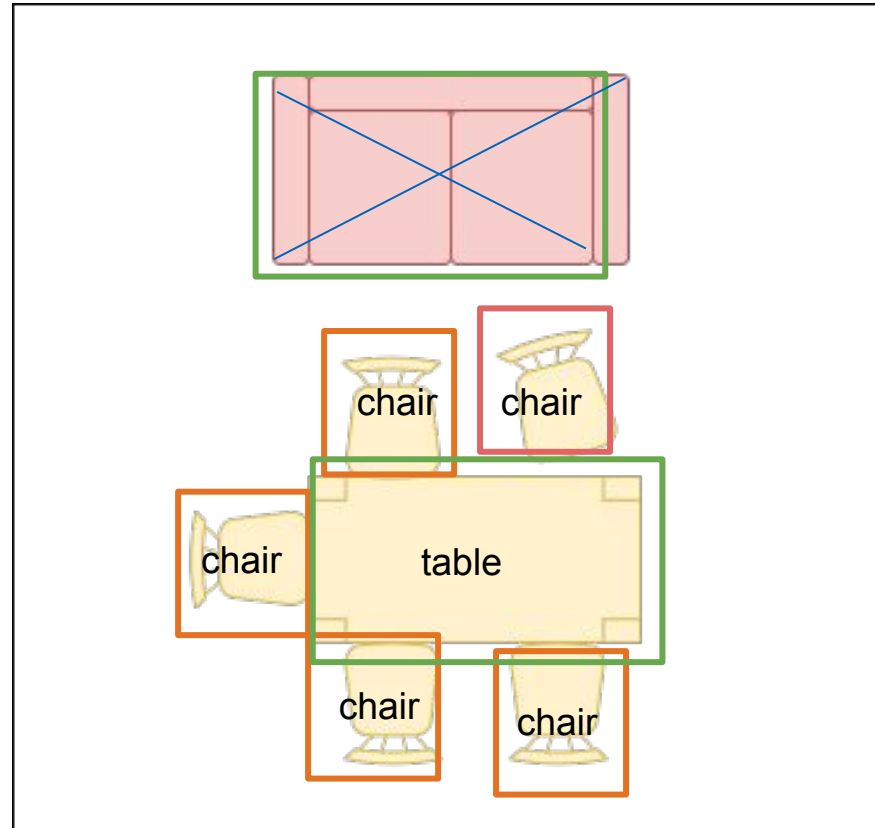




- Gather the MC samples
- Compute epistemic uncertainty

Method: Improving Detection Performance

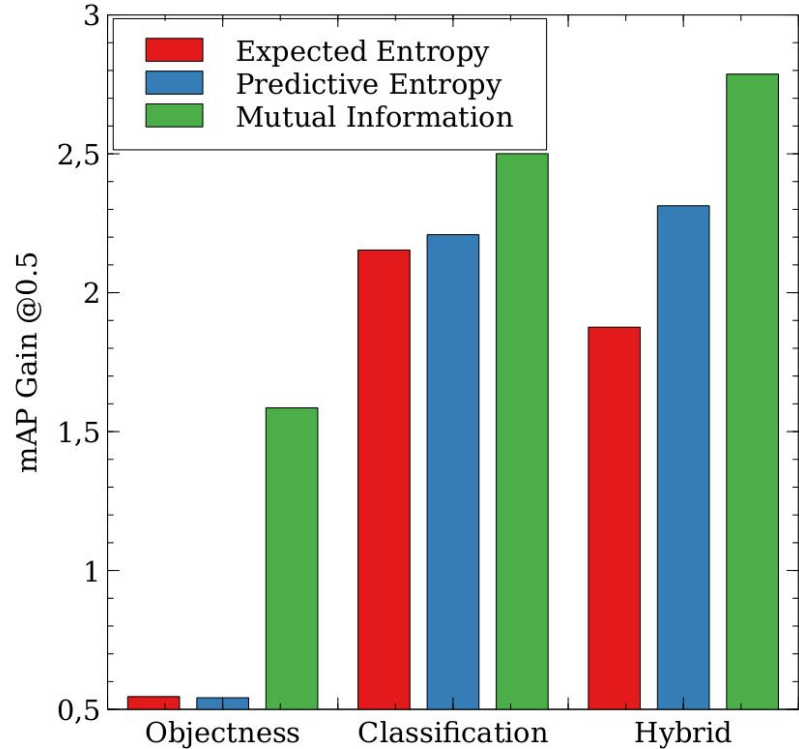
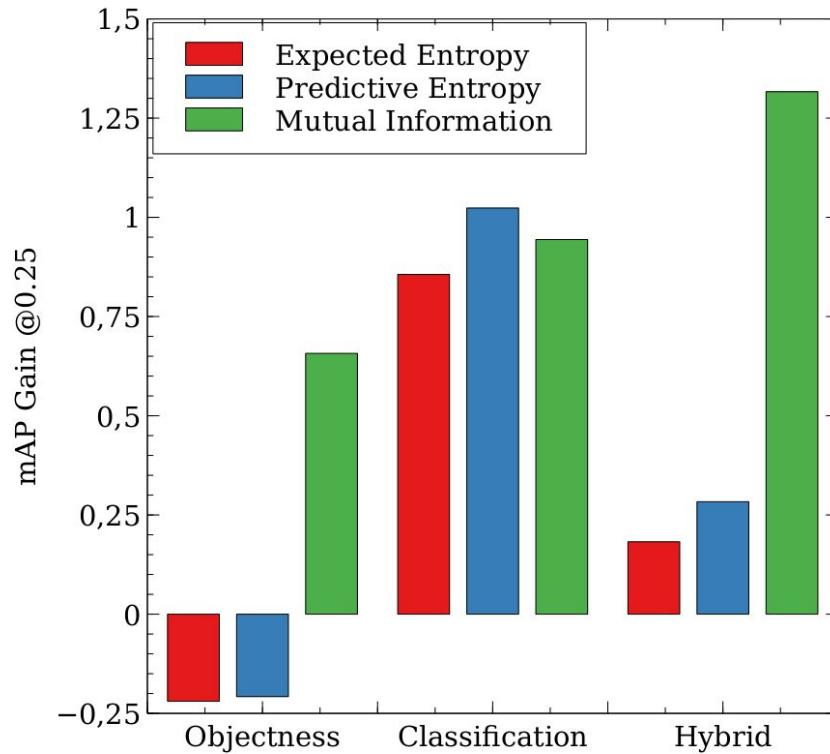
Use uncertainty as
a weight for the
proposal



Network is not
sure!

Rejected!

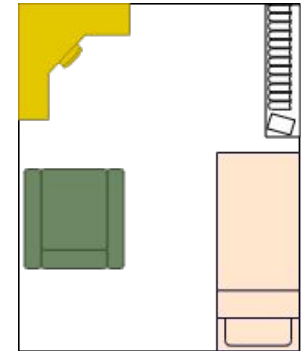
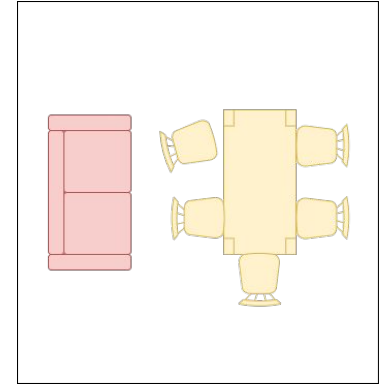
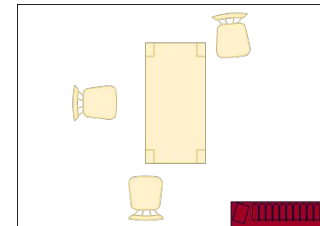
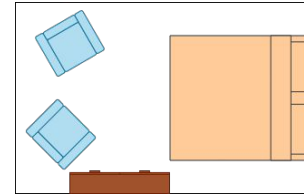
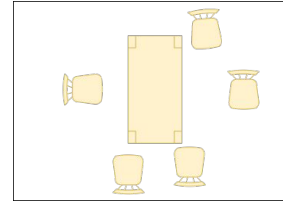
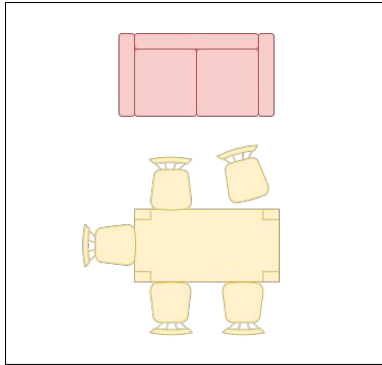
Results: Performance Boost in Detections



Method: Active Learning

Unlabeled data

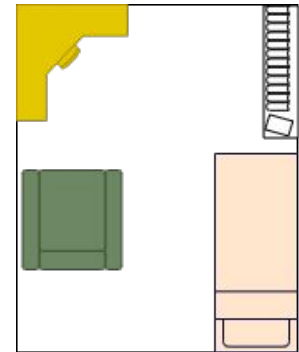
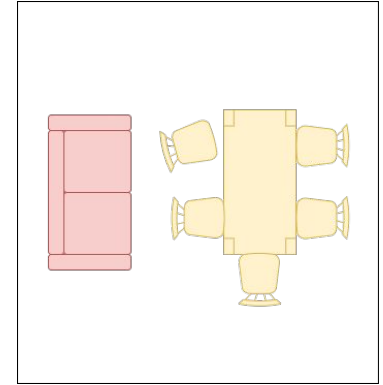
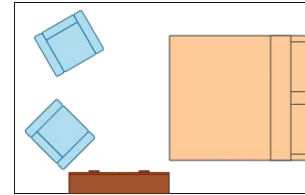
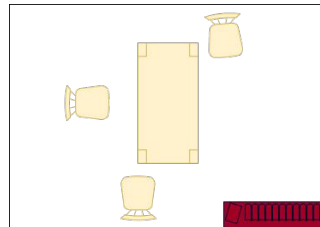
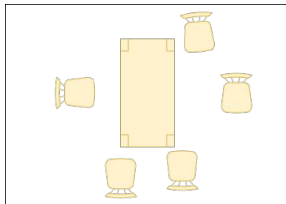
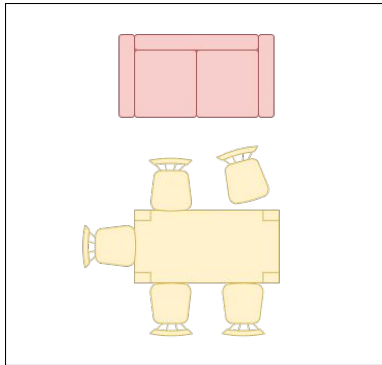
Labeled Train Set



Method: Random Selection

Unlabeled data

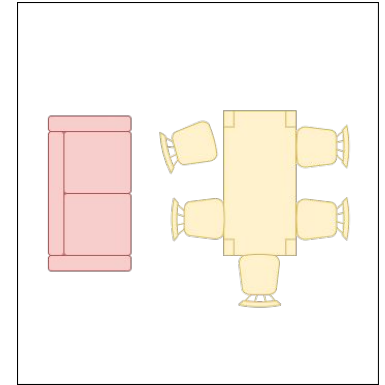
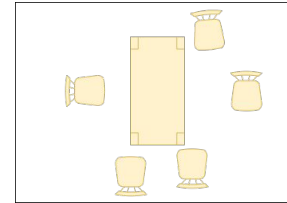
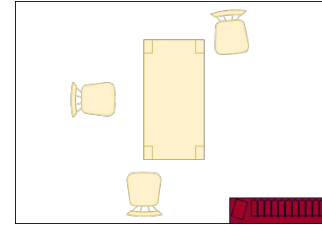
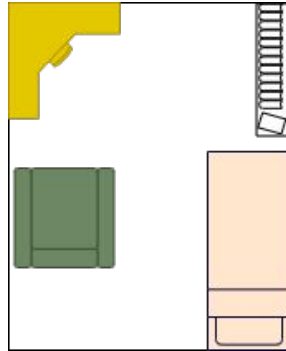
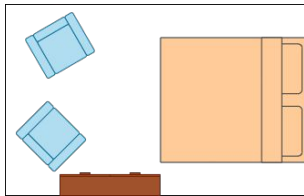
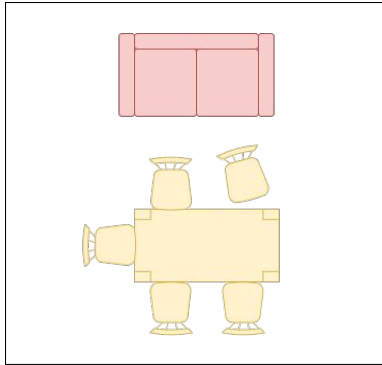
Labeled Train Set



Method: Uncertainty Selection

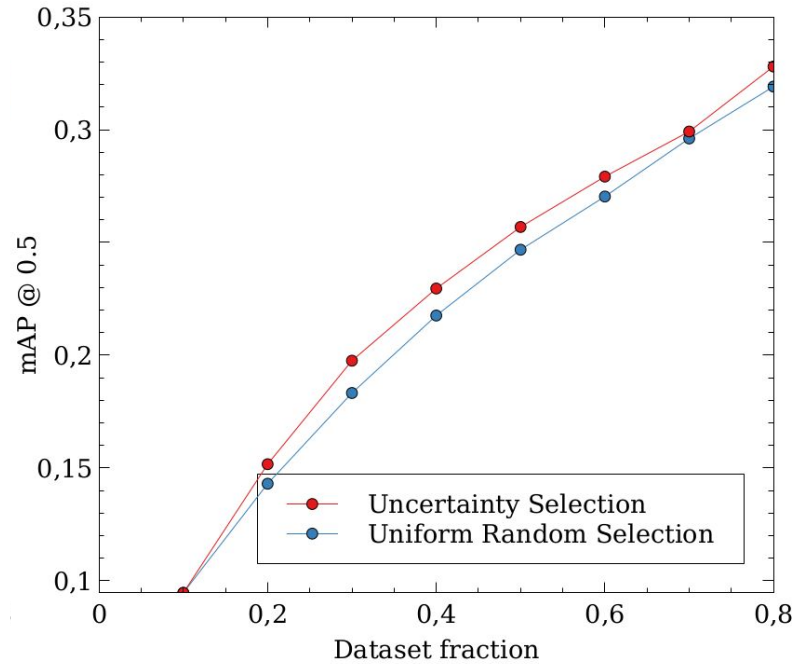
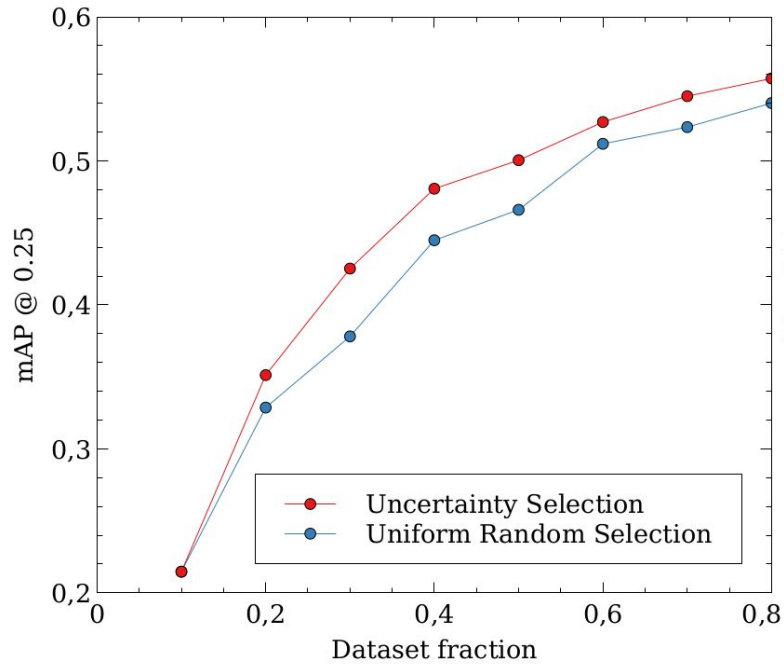
Unlabeled data

Labeled Train Set



Select samples with high epistemic uncertainty!

Results: Active Learning



Conclusion & Future Work

- Even with simple techniques, we are able get visible improvements
- More complex uncertainty methods to be explored
- Aleatoric uncertainty in point clouds
- More advanced active learning schemes to be tried
- More complex weighting schemes can be tried

Appendix

Uncertainty Type	mAP @0.25	mAP @ 0.5
Native	0.57087	0.3339318
Objectness	0.57647	0.3449952
Classification	0.5795556	0.3570348
Hybrid	0.5848642	0.3620454

Uncertainty Type	AR @0.25	AR @ 0.5
Native	0.57087	0.4993466
Objectness	0.57647	0.5056164
Classification	0.5795556	0.5171316
Hybrid	0.5848642	0.5173844

Appendix

Dropout probability	mAP @ 0.25	mAP @ 0.5
p=0	57	35
p=0.1	57.5	34.5
p=0.2	55.5	30
p=0.3	51	23
p=0.5	46	15

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